CHAPTER-1
INTRODUCTION

1.1 Overview of the Project

Natural disasters such as floods, landslides, earthquakes, and forest fires have become increasingly frequent and severe, causing significant damage to human life, infrastructure, and the environment. Effective disaster management requires timely detection and rapid response to minimize these impacts. However, many existing systems depend heavily on stable internet connectivity and centralized infrastructure, which are often unavailable in remote or rural areas.

This project presents a LoRa-based Smart Disaster Prediction and Alert System that integrates Internet of Things (IoT) devices and Artificial Intelligence (AI) techniques. The system continuously monitors environmental parameters such as temperature, humidity, water level, and gas concentration using sensors. These parameters are transmitted over long distances using LoRa (Long Range) communication, which operates efficiently without internet connectivity. The collected data is analyzed to detect abnormal conditions and generate early warnings, enabling timely action and improved disaster preparedness.

1.2 Motivation of the Project

The motivation behind this project arises from the increasing impact of natural disasters and the limitations of existing monitoring systems. In many disaster-prone regions, especially rural and remote areas, there is a lack of reliable infrastructure for continuous monitoring and communication. This leads to delayed warnings and increased risk to human life and property.

Additionally, most traditional systems focus only on detecting disasters after they occur rather than predicting them in advance. There is a growing need for intelligent systems that can analyze environmental data and provide early warnings. This project is motivated by the goal of developing a low-cost, reliable, and intelligent solution that can operate independently of internet connectivity and provide real-time disaster alerts.

1.3 Problem Statement

Despite advancements in technology, existing disaster monitoring systems face several challenges. These include dependency on internet connectivity, limited communication range, high power consumption, and lack of predictive capabilities. As a result, these systems are not suitable for deployment in remote areas where disasters often occur.

The problem addressed in this project is the lack of an efficient, reliable, and predictive disaster monitoring system that can operate in low-connectivity environments. There is a need for a system that can continuously monitor environmental conditions, transmit data over long distances, and intelligently predict potential disasters to provide timely alerts.

1.4 Project Objectives

The primary objective of this project is to design and develop a smart disaster prediction and alert system using IoT, LoRa, and AI technologies. The system aims to continuously monitor environmental parameters using sensors and transmit the collected data over long distances using LoRa communication.

Another objective is to incorporate machine learning techniques to analyze the collected data and predict potential disaster conditions. The system is also designed to generate real-time alerts through multiple channels, ensuring timely communication to authorities and users.

Additionally, the project aims to develop a low-power, scalable, and cost-effective solution that can be deployed in rural and remote areas. Ultimately, the objective is to reduce disaster impact by improving early warning systems and enhancing response time.

1.5 Scope and Limitations

Scope:

The proposed system is designed for monitoring environmental conditions and detecting potential disasters such as floods, fires, and landslides. It can be deployed in rural areas, forest regions, industrial zones, and disaster-prone locations. The system supports real-time data collection, long-range communication, and early warning alerts.

It also provides flexibility for future expansion by allowing the integration of additional sensors and advanced data analysis techniques. The system can be scaled to cover larger geographical areas by deploying multiple sensor nodes.

Limitations:

Despite its advantages, the system has certain limitations. The accuracy of predictions depends on the quality and calibration of sensors. Environmental factors such as extreme weather conditions may affect sensor performance. Additionally, the AI model used in the system may require further training with large datasets to improve prediction accuracy.

The system is also limited by hardware constraints such as battery life and processing capability. While LoRa provides long-range communication, it has a lower data transmission rate compared to other communication technologies.

1.6 Alignment with SDGs

The proposed project aligns with several United Nations Sustainable Development Goals (SDGs) by contributing to disaster resilience and environmental sustainability.

Table 1.1: Contribution of the Project to SDGs

SDG Goal	Title	Contribution of the Project
SDG 9	Industry, Innovation, and Infrastructure	The project promotes the use of advanced technologies such as IoT, AI, and LoRa to build reliable and innovative infrastructure for disaster monitoring.

SDG 11	Sustainable Cities and Communities	It enhances disaster preparedness and improves safety in both urban and rural areas through early warning systems.
SDG 13	Climate Action	The system helps in monitoring environmental changes and supports timely action against climate-related disasters.
SDG 3	Good Health and Well-being	By providing early alerts, the system reduces risks to human life and supports emergency response efforts.

1.7 Organization of Report

This report is organized into multiple chapters to provide a clear understanding of the project.

- **Chapter 1 (Introduction):** Presents the overview, motivation, problem statement, objectives, scope, and alignment with SDGs.
- **Chapter 2 (Literature Survey):** Discusses existing research and related work in disaster monitoring systems.
- **Chapter 3 (Proposed System):** Describes the system architecture, components, and design methodology.
- **Chapter 4 (Technologies Used – IoT and AI):** Explains the role of IoT in real-time monitoring and introduces Artificial Intelligence concepts.
- **Chapter 5 (Result and Analysis):** Presents the outcomes and performance analysis of the system.
- **Chapter 6 (Conclusion and Future Scope):** Summarizes the project and suggests possible improvements and future enhancements.

CHAPTER-2

LITERATURE SURVEY

2.1 Review of Existing System

Recent research shows that LoRa and IoT-based systems play a crucial role in disaster monitoring and early warning due to their long-range communication, low power consumption, and independence from internet infrastructure [10][12][15].

A study by Melvin P. Manuel et al. (2024) proposed a LoRa-based disaster management system for search and rescue missions, highlighting the importance of reliable communication in damaged environments. The system used robotic coordination and LoRa networks to maintain communication between rescue teams and base stations, improving efficiency in disaster response [1].

Similarly, Shubhi Agrawal et al. (2024) developed a real-time disaster response system using multiple sensors integrated with LoRa technology. Their work emphasized the importance of multi-sensor data collection for accurate disaster detection and faster emergency response [2].

Research by Lopes et al. (2025) evaluated the performance of LoRa networks in disaster monitoring scenarios and identified challenges such as security vulnerabilities and network reliability, which must be addressed for large-scale deployment [4].

Another study presented a low-power LoRa-based disaster alert system specifically designed for remote areas, proving that such systems are highly effective where communication infrastructure is unavailable [5].

Shinde et al. (2025) proposed an IoT-based automatic disaster monitoring system using LoRa and multiple sensors to detect parameters such as temperature, gas leakage, and flood levels. The system demonstrated real-time monitoring and alert generation capabilities [6].

Syed et al. (2024) analyzed the feasibility of LoRa-based communication systems in disaster scenarios and concluded that LoRa mesh networks can significantly improve connectivity and coverage in affected regions [7].

Ranasinghe et al. (2024) introduced a multi-hop LoRa architecture for earthquake early warning systems, showing improved communication reliability without relying on the internet [8].

Another important contribution is the stochastic modeling of IoT disaster detection systems, which helps optimize system performance and reliability in different geographic regions [11].

Additional studies highlight scalable LoRaWAN architectures, seamless long-range communication, and smart city disaster resilience frameworks, further strengthening the applicability of LoRa in disaster management systems [12][13][14].

Overall, the literature indicates that LoRa-based disaster systems are cost-effective, scalable, and highly suitable for remote monitoring, but future improvements are needed in security, scalability, and intelligent prediction using AI/ML techniques [3][9][15].

2.2 Limitations of Existing Approaches

Although existing systems contribute significantly to disaster monitoring, they exhibit several limitations that reduce their overall effectiveness.

A major limitation is the dependency on internet connectivity. Many IoT-based systems rely on cloud platforms for data processing and communication, making them unsuitable for areas with poor or no network coverage. This results in delays in data transmission and alert generation during critical situations.

Another issue is the lack of long-range communication capability. Technologies such as Wi-Fi and Bluetooth have limited range, which restricts their use in large-scale monitoring systems. As a result, these systems cannot effectively cover wide geographical areas without additional infrastructure.

Many existing solutions also suffer from the absence of predictive capabilities. Most systems are designed to detect disasters only after certain thresholds are exceeded, rather than predicting them in advance. This reactive approach reduces the time available for taking preventive actions.

High power consumption is another significant limitation, especially in remote deployments where continuous power supply is not guaranteed. Systems that require frequent maintenance or battery replacement are not practical for long-term use.

Furthermore, there is often a lack of integration between sensing, communication, and data analysis components. Many systems focus on only one aspect, such as monitoring or communication, without providing a complete and unified solution.

2.3 Need for Proposed System

The limitations of existing approaches highlight the need for a more efficient and integrated disaster monitoring solution. There is a clear requirement for a system that can operate reliably in low-connectivity and remote environments without depending on internet infrastructure.

An effective system should incorporate long-range communication technology to ensure seamless data transmission over large distances. It should also be capable of real-time monitoring using multiple sensors, allowing accurate observation of environmental conditions.

In addition, the inclusion of intelligent data analysis using AI and machine learning is essential to move from reactive detection to proactive prediction. This enables early identification of potential disaster scenarios and provides more time for preventive measures.

There is also a need for a system that is energy-efficient, cost-effective, and scalable, so that it can be deployed across multiple locations with minimal maintenance. Such a system should integrate sensing, communication, and alert mechanisms into a single framework to ensure reliability and ease of operation.

The proposed LoRa-based Smart Disaster Prediction and Alert System address these requirements by combining IoT sensors, long-range communication, and AI-based prediction. This integrated approach overcomes the limitations of existing systems and provides a more robust and practical solution for disaster management.

CHAPTER-3
PROPOSED SYSTEM

3.1 Introduction

The proposed system is designed to provide an efficient and intelligent solution for disaster prediction and alerting by integrating IoT sensors, LoRa communication, and embedded processing. The system focuses on continuous environmental monitoring and early detection of abnormal conditions that may lead to disasters such as floods, fires, and gas leaks.

Unlike traditional systems that depend on internet connectivity, the proposed system utilizes LoRa technology, which enables long-range communication with low power consumption. This makes the system highly suitable for deployment in rural and remote areas. The system is capable of detecting critical environmental changes and generating immediate alerts, thereby improving disaster response time and reducing potential damage.

3.2 System Architecture

The system architecture consists of two main units:

1. Transmitter Unit (Sensor Node)
2. Receiver Unit (Alert Node)

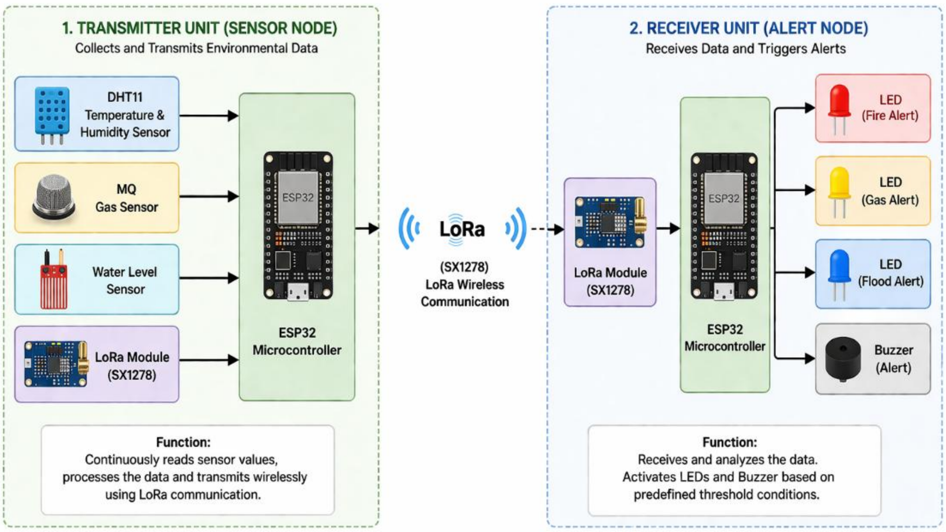

Fig 3.1. Lora Transmission System Architecture

1. Transmitter Unit (Sensor Node)

The transmitter unit is responsible for collecting environmental data. It includes:

- ESP32 microcontroller
- DHT11 temperature and humidity sensor
- MQ gas sensor
- Water level sensor
- LoRa module (SX1278)

This unit continuously reads sensor values, processes them, and transmits the data wirelessly using LoRa communication.

2. Receiver Unit (Alert Node)

The receiver unit is responsible for receiving and analyzing the transmitted data. It includes:

- ESP32 microcontroller
- LoRa module (SX1278)
- LEDs for indication (fire, gas, flood)
- Buzzer for alert

The receiver processes incoming data and activates alerts based on predefined threshold conditions.

The overall system forms a wireless monitoring network, where the transmitter collects data and the receiver provides alerts in real time.

3.3 Components Used in Proposed System

3.3.1 ESP32 Microcontroller

Fig 3.2. ESP32 Microcontroller

The ESP32 is the main processing unit of the system. It is used in both transmitter and receiver sections. It collects sensor data, processes it, and controls communication between devices. The ESP32 is chosen due to its high performance, low power consumption, and multiple input/output pins.

Principle: Digital processing + embedded computing

- Based on a microcontroller architecture (CPU + memory + GPIO).
- Works on binary logic (0s and 1s) and executes programmed instructions.
- Uses ADC (Analog to Digital Conversion) to read sensor data.
- Controls devices using digital output signals.

3.3.2 LoRa Module (SX1278 / RA-02)

Fig 3.3. 2 LoRa Module (SX1278 / RA-02)

The LoRa module enables long-range wireless communication between the transmitter and receiver units. It operates on low power and can transmit data over several kilometers. It is ideal for remote monitoring applications where internet connectivity is not available.

Principle: Chirp Spread Spectrum (CSS) modulation

- Uses radio frequency (RF) communication.
- Applies spread spectrum technique → signals are spread over a wide frequency band.
- This makes it:
  - Long-range
  - Low power
  - Resistant to noise

3.3.3 DHT11 Temperature and Humidity Sensor

Fig 3.4. DHT11 Sensor

The DHT11 sensor is used to measure temperature and humidity in the environment. It helps in detecting abnormal temperature rise, which can indicate fire or overheating conditions.

Principle:

- Humidity: Capacitive sensing
- Temperature: Thermistor (resistance change)
- Humidity changes → capacitance changes
- Temperature changes → resistance changes

3.3.4 MQ Gas Sensor

Fig 3.5. MQ Gas Sensor

The MQ gas sensor is used to detect the presence of harmful gases or smoke in the environment. It plays a crucial role in identifying gas leaks or fire hazards.

Principle: Chemiresistive sensing

- Uses tin dioxide (SnO_2) material.
- When gas is present → resistance changes.
- More gas concentration → larger change in resistance.

3.3.5 Water Level Sensor

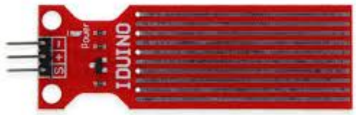

Fig 3.6. Water Level Sensor

The water level sensor is used to monitor the level of water in a specific area. It helps in detecting flood conditions when the water level exceeds a predefined threshold.

Principle: Electrical conductivity

- Water conducts electricity.
- When water touches probes → current flows.
- Higher water level → more conductive paths.

3.3.6 LEDs (Red, Yellow, Blue)

Fig 3.7. LEDs

LEDs are used for visual indication of different types of disasters:

- Red LED: Fire alert
- Yellow LED: Gas leak alert
- Blue LED: Flood alert

They provide quick and clear visual warnings.

Principle: Electroluminescence

- Based on p-n junction semiconductor.
- When current flows → electrons recombine → light is emitted.

3.3.7 Buzzer

Fig 3.8. Buzzer

The buzzer is used to generate an audible alert during emergency situations. It ensures that warnings are noticeable even if the user is not observing the display or LEDs.

Principle: Piezoelectric effect

- Electric signal → mechanical vibration → sound.
- Vibrating plate produces audible alert.

3.3.8 Power Supply

The system is powered using a stable power source, USB supply. Proper voltage regulation is maintained to ensure safe operation of components like ESP32 and LoRa module.

3.4 Working Principle

The working of the proposed system is based on continuous sensing, wireless communication, and alert generation.

At the transmitter side, the ESP32 reads data from the DHT11 sensor, MQ gas sensor, and water level sensor. These sensors continuously monitor environmental conditions such as temperature, gas concentration, and water levels. The collected data is processed and transmitted using the LoRa module.

At the receiver side, another ESP32 receives the transmitted data through the LoRa module. The received values are compared with predefined threshold levels to determine whether a disaster condition exists.

Table 3.1: Alert Conditions

Condition	Sensor Value	Output
Normal	Within limits	No alert
Fire	Temp > 40°C	Red LED + Buzzer
Gas	Gas > Threshold	Yellow LED + Buzzer
Flood	Water level high	Blue LED + Buzzer

The system operates under the following conditions:

- **Normal Condition:** All sensor values remain within safe limits. No alerts are generated, and all LEDs and buzzer remain OFF.
- **Fire Alert:** When temperature exceeds a certain threshold (Example, 40°C), the system identifies a fire risk. The red LED is activated, and the buzzer is turned ON.
- **Gas Leak Alert:** When gas concentration exceeds the predefined threshold, the system detects a gas leak. The yellow LED is activated along with the buzzer.
- **Flood Alert:** When the water level rises above the safe limit, the system identifies a flood condition. The blue LED is activated, and the buzzer is triggered.

CHAPTER-4

TECHNOLOGIES USED (IoT AND AI)

4.1 Introduction to IoT

The Internet of Things (IoT) refers to a network of connected devices that collect and exchange data in real time using sensors and communication technologies. It enables automatic monitoring without human intervention.

In disaster management, IoT is used to continuously monitor environmental parameters such as temperature, gas levels, and water levels. This helps in detecting abnormal conditions early and enables quick response to potential disasters.

4.2 IoT in Proposed System

In the proposed system, the Internet of Things (IoT) is used to enable real-time monitoring of environmental conditions through connected sensors. Devices such as the DHT11 sensor, MQ gas sensor, and water level sensor are interfaced with the ESP32 microcontroller to continuously collect data related to temperature, gas concentration, and water levels.

The ESP32 processes the collected sensor data and transmits it to the receiver unit using LoRa communication. This ensures that data can be sent over long distances without relying on internet connectivity. The continuous data flow allows the system to monitor environmental changes in real time and detect abnormal conditions.

Thus, IoT in this project plays a key role in data collection, communication, and real-time monitoring, forming the foundation for effective disaster detection and alert generation.

4.3 Introduction to Artificial Intelligence (AI)

Artificial Intelligence (AI) refers to the ability of machines or computer systems to perform tasks that normally require human intelligence, such as decision-making, pattern recognition, and prediction.

In simple terms, AI helps systems to analyze data and make smart decisions automatically. Instead of only detecting current conditions, AI can study patterns in data and predict what might happen next.

In disaster management, AI is used to analyze environmental data collected from sensors and identify abnormal patterns that may indicate a possible disaster. This helps in providing early warnings and improving the accuracy of the system.

In this project, AI concepts such as Decision Tree and Random Forest can be used to improve prediction by analyzing sensor data and identifying potential disaster conditions more effectively.

4.4 Decision Tree Algorithm

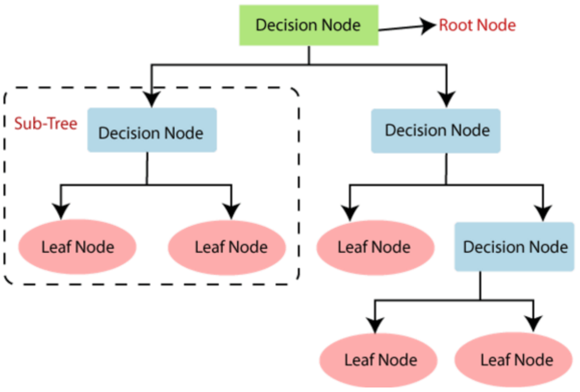

Fig 4.1. Decision Tree Algorithm

A Decision Tree is a simple and intuitive machine learning algorithm used for making decisions based on a set of conditions. It follows a tree-like structure, where each node represents a condition, each branch represents the outcome, and the leaf node represents the final decision or classification.

The algorithm works by splitting the data into smaller parts based on certain conditions. These conditions are usually based on threshold values of input features. For example, in a disaster monitoring system:

- If temperature is high → possible fire
- If gas level is high → gas leak
- If water level is high → flood

The system evaluates these conditions step-by-step, moving through the tree until it reaches a final decision. This stepwise process makes the Decision Tree easy to understand and interpret.

One of the key advantages of a Decision Tree is that it can handle multiple input parameters at once and clearly show how decisions are made. It does not require complex mathematical calculations, making it suitable for simple and real-time applications.

In addition, Decision Trees can be trained using historical data, where the system learns patterns between input values (sensor data) and output labels (type of disaster). Once trained, the model can quickly classify new incoming data.

However, a single Decision Tree may sometimes produce less accurate results if the data is complex, as it can be sensitive to small changes in input values.

In this project, a Decision Tree can be used to analyze sensor data from temperature, gas, and water sensors to classify whether the condition is normal or a specific type of disaster. It helps in making quick and understandable decisions based on real-time environmental data.

4.5 Random Forest Algorithm

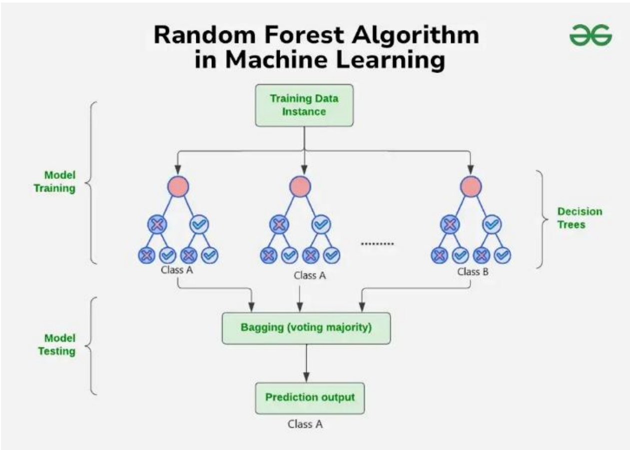

Fig 4.2. Random Forest Algorithm

Random Forest is an advanced machine learning algorithm that uses multiple Decision Trees to make more accurate predictions. Instead of relying on a single tree, it combines the results of many trees and gives the final decision based on majority voting.

This approach improves accuracy and reduces errors compared to a single Decision Tree. It is more reliable because it considers multiple possibilities before making a decision.

In this project, Random Forest can be used to analyze sensor data more effectively and provide better prediction of disaster conditions, reducing false alarms and improving system performance.

4.6 Classification (Neural Network - NN)

Classification using Neural Networks is a machine learning approach where the model learns patterns from input data and assigns it to specific categories.

- Uses layers of interconnected neurons to process data
- Takes sensor inputs (temperature, gas, water level)
- Learns patterns to classify conditions (Normal / Fire / Gas / Flood)
- Can improve prediction accuracy compared to simple threshold methods

Neural Networks consist of input, hidden, and output layers, where each layer processes data and passes it forward. During training, the model adjusts its internal weights using algorithms such as backpropagation to improve accuracy. This allows the system to handle complex relationships between multiple sensor parameters and make intelligent predictions.

It can be used to predict disaster type based on sensor data.

4.7 Keras

Keras is a high-level deep learning library used to build and train neural networks easily.

- Built on top of TensorFlow
- Simple and user-friendly API
- Used for creating Neural Network models
- Supports layers, activation functions, and training

Can be used to build and train NN models for classification.

4.8 EON Tuner

EON Tuner is a tool (used in platforms like Edge Impulse) that optimizes machine learning models for embedded systems.

- Automatically selects best model architecture
- Optimizes for speed, accuracy, and memory
- Designed for low-power devices like microcontrollers
- Helps deploy ML models on edge devices

EON Tuner evaluates multiple model configurations and selects the most efficient one based on performance constraints. It reduces model size and computational requirements, enabling deployment on resource-limited devices. This makes it highly suitable for real-time applications where low latency and power efficiency are important.

It is useful for running AI models efficiently on ESP32.

4.9 Role of AI in Proposed System

Artificial Intelligence (AI) plays an important role in enhancing the capability of the proposed disaster monitoring system by enabling intelligent analysis and prediction of environmental conditions. Instead of only reacting to threshold values, AI helps the system to understand patterns in sensor data and make better decisions.

In the proposed system, AI algorithms such as Decision Tree and Random Forest can be used to analyze data collected from sensors like temperature, gas, and water level sensors. These algorithms process the data and identify whether the conditions indicate normal operation or a potential disaster.

The use of AI improves the system in several ways:

- **Early Prediction:** AI can identify patterns that indicate a possible disaster before it actually occurs.
- **Improved Accuracy:** By analyzing multiple parameters together, AI reduces incorrect alerts.
- **Reduced False Alarms:** Intelligent decision-making helps avoid unnecessary warnings.
- **Better Decision Support:** Provides more reliable information for taking preventive actions.

Currently, the system may use simple threshold-based logic, but AI can be integrated in future to make the system more advanced and efficient.

Thus, AI adds intelligence to the system, making it more reliable, accurate, and capable of handling real-world disaster scenarios.

4.10 Edge Impulse-Based Machine Learning Model

To improve the intelligence of the proposed system, an Edge Impulse based machine learning model was trained to classify sensor patterns and support early disaster prediction. The model uses environmental readings from the DHT11 temperature and humidity sensor, MQ gas sensor, and water level sensor to identify whether the situation is normal or corresponds to a fire/overheat, flood, or gas leak condition.

The classifier was prepared for embedded deployment, and the selected model version is **Quantized (int8)**. Quantization reduces model size and improves execution speed, making the model suitable for real-time inference on resource-constrained hardware such as the ESP32.

4.10.1 Model Configuration

Parameter	Details
Development platform	Edge Impulse
Model type	Multi-class classification
Deployment format	Quantized (int8)
Target device	ESP32-based embedded system
Input source	Sensor readings from temperature, gas, and water level modules
Output classes	4 classes: Fire / Overheat, Flood, Gas Leak, Normal

4.10.2 Class Labels Used by the Model

Class Name	Meaning in the System
FIRE / OVERHEAT ALERT!	Possible fire or overheating condition
FLOOD ALERT!	Possible flood condition
GAS LEAK ALERT!	Possible gas leak condition
NORMAL	Safe or normal environmental condition

The trained model learns patterns from labeled sensor data and associates them with the corresponding disaster category. This allows the system to move beyond fixed threshold logic and supports smarter decision-making at the edge. The low-latency quantized model is especially useful in this project because it can run close to the sensor node and generate a prediction quickly before an alert is issued.

CHAPTER-5
RESULTS AND ANALYSIS

5.1 Technologies and Tools Used

The successful implementation of the proposed system is achieved through the integration of multiple technologies and development tools. The system primarily utilizes Internet of Things (IoT) technology for real-time data collection through sensors, enabling continuous environmental monitoring. For communication, LoRa (Long Range) technology is used to transmit data between the transmitter and receiver units. LoRa provides reliable long-distance communication with low power consumption, making it suitable for remote and disaster-prone areas. The system is built around embedded systems design, with the ESP32 microcontroller acting as the core processing unit, handling sensor data acquisition, processing, and communication tasks. On the software side, the system is developed using the Arduino IDE, which provides a simple and efficient environment for programming the ESP32, while Embedded C is used for writing the control logic, including sensor interfacing, data transmission, and alert generation.

Additionally, Edge Impulse is used for developing and optimizing machine learning models for sensor data analysis, enabling efficient deployment of AI models on the ESP32. Serial communication tools are also used for debugging and monitoring real-time data during system testing. These technologies collectively ensure efficient operation, reliability, and ease of implementation.

5.2 Hardware Specifications

The hardware components used in the system are selected based on performance, compatibility, and power efficiency.

- ESP32 Microcontroller
- LoRa Module (SX1278)
- DHT11 Sensor
- MQ Gas Sensor
- Water Level Sensor
- LED Indicators
- Buzzer
- Power Supply

5.3 Software Specifications

The software implementation plays a crucial role in controlling the system and processing sensor data.

- **Programming Language:** Embedded C is used to write the control logic for the system.
- **Development Environment:** Arduino IDE is used for coding, compiling, uploading programs to the ESP32, and edge impulse for training machine learning model.
- **Libraries Used:** Standard libraries for ESP32, LoRa communication, Edge impulse based library and sensor interfacing (such as DHT library) are used to simplify development.
- **Functional Modules:**
    - Sensor data acquisition
    - Data processing and threshold comparison
    - LoRa data transmission and reception
    - Alert generation logic
- **Serial Monitor:** Used for displaying real-time sensor values and debugging during testing. The software is designed to ensure continuous monitoring, efficient data handling, and reliable alert generation.

5.4 Experimental Results

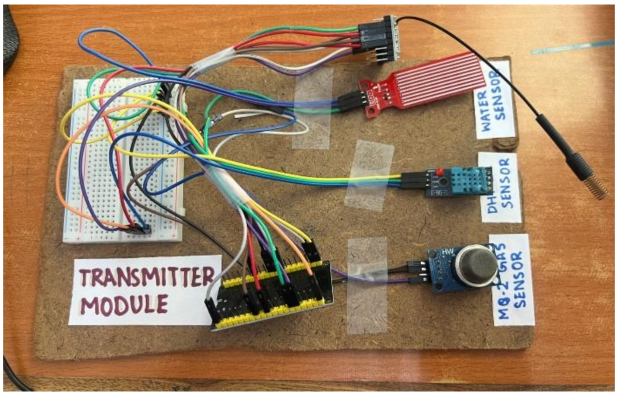

Fig 5.1. Transmitter Module

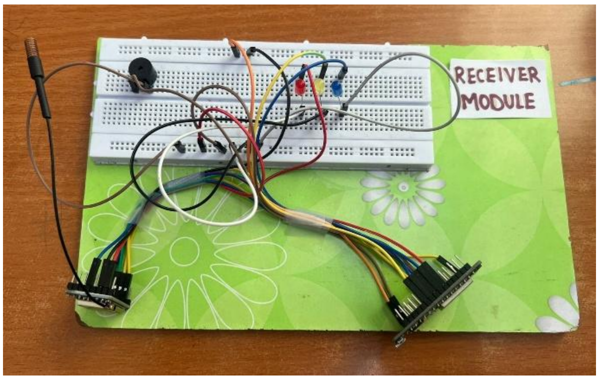

Fig 5.2. Receiver Module

The system was tested under different environmental conditions to evaluate its performance and functionality.

During normal conditions, the sensor values remained within safe limits, and no alerts were generated. All LEDs remained OFF, and the buzzer was inactive, indicating stable system behaviour.

When the temperature was increased beyond the predefined threshold (approximately 40°C), the system successfully detected a potential fire condition. The red LED was activated, and the buzzer produced an audible alert, confirming proper operation.

Similarly, when gas concentration levels increased, the system identified a gas leak condition. The yellow LED was turned ON, and the buzzer was activated, demonstrating accurate detection.

For flood simulation, water was introduced to the water level sensor. Once the water level exceeded the threshold, the system triggered the blue LED along with the buzzer, indicating a flood alert.

In all cases, the transmitter successfully sent data via LoRa, and the receiver accurately processed the received data and generated appropriate alerts. The system demonstrated reliable communication and quick response to changing conditions.

5.5 Performance Evaluation

The performance of the system was evaluated based on key parameters such as accuracy, response time, communication reliability, and power efficiency.

- **Accuracy:** The system provided accurate detection of environmental changes based on sensor inputs. However, accuracy is dependent on sensor quality and calibration.

- **Response Time:** The system responded quickly to threshold changes, generating alerts almost immediately after detecting abnormal conditions.
- **Communication Reliability:** LoRa communication proved to be stable and reliable over long distances, even without internet connectivity. Data transmission was consistent with minimal loss.
- **Power Efficiency:** The system consumed low power, making it suitable for battery-operated and long-term deployment.
- **Scalability:** The modular design allows easy addition of more sensor nodes, enabling expansion to larger monitoring areas.

Overall, the system demonstrated effective performance in real-time monitoring, reliable communication, and timely alert generation, making it a suitable solution for disaster prediction and management.

5.6 Machine Learning Model Performance Analysis

The Edge Impulse classifier was validated on **85** samples. The results show strong overall performance and indicate that the model is suitable for edge deployment in the disaster prediction system.

5.6.1 Overall Validation Metrics

Metric	Value
Validation accuracy	97.6%
Loss	0.16
Area under ROC curve	1.00
Weighted precision	0.98
Weighted recall	0.98
Weighted F1-score	0.98

5.6.2 Confusion Matrix (Validation Set)

Actual \ Predicted	Fire / Overheat	Flood	Gas Leak	Normal
Fire / Overheat	15	0	0	1
Flood	0	4	0	0
Gas Leak	0	0	10	0
Normal	1	0	0	54

5.6.3 Class-Wise Performance

Class	Precision	Recall	F1-score
Fire / Overheat	0.9375	0.9375	0.9375
Flood	1.0000	1.0000	1.0000
Gas Leak	1.0000	1.0000	1.0000
Normal	0.9818	0.9818	0.9818

The confusion matrix shows that the model correctly classified **83 out of 85** validation samples. Only two samples were misclassified: one fire/overheat sample was predicted as normal, and one normal sample was predicted as fire/overheat. Even with these minor errors, the model achieved a high validation accuracy of **97.6%** and a near-perfect ROC-AUC of approximately **1.00**. These results confirm that the quantized Edge Impulse model is highly effective for real-time disaster classification in the proposed system.

CHAPTER-6
CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

The LoRa-based Smart Disaster Prediction and Alert System provides an efficient and practical solution for disaster monitoring and early warning. It integrates IoT sensors, long-range communication, and embedded systems to detect hazards like floods, fires, and gas leaks. The use of LoRa enables reliable long-distance data transmission without internet, making it suitable for remote areas. The system effectively detects abnormal conditions and generates timely alerts while maintaining low power consumption. Overall, it is a cost-effective, scalable, and reliable solution that enhances safety and supports quick decision-making during emergencies.

6.2 Future Scope

The system can be improved by incorporating AI and Machine Learning for more accurate disaster prediction beyond simple threshold detection. Adding advanced sensors such as seismic and soil moisture sensors can expand its capability to monitor more disaster types, making the system more versatile and intelligent.

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